

Semântica Operacional Big-Step de Uma Linguagem Imperativa

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1 Semântica Operacional Big-Step da Linguagem IMP

1.1 Sintaxe

$n \in \mathbb{N}$	números naturais
$x \in \mathbf{Var}$	variáveis
$b \in \{\text{True}, \text{False}\}$	valores booleanos
E	expressões aritméticas
$E ::= n$	número inteiro
$ x$	variável
$ E + E$	soma
$ E * E$	multiplicação
B	expressões booleanas
$B ::= b$	valor booleano
$ \neg B$	negação
$ B \vee B$	ou
$ B \wedge B$	e
$ E \leq E$	menor ou igual
C	comandos
$C ::= \text{Skip}$	skip
$ x := E$	atribuição
$ C ; C$	sequência de comandos
$ \text{if } B \text{ then } C \text{ else } C$	if
$ \text{while } B \text{ do } C$	while

1.2 Semântica de Expressões Aritméticas

$$\frac{}{\langle n, \sigma \rangle \Downarrow n} \text{ (NUM)}$$

$$\frac{}{\langle x, \sigma \rangle \Downarrow \sigma(x)} \text{ (VAR)}$$

$$\frac{\langle E_1, \sigma \rangle \Downarrow n_1 \quad \langle E_2, \sigma \rangle \Downarrow n_2}{\langle E_1 + E_2, \sigma \rangle \Downarrow n} \quad n = n_1 + n_2 \text{ (SOMA)}$$

$$\frac{\langle E_1, \sigma \rangle \Downarrow n_1 \quad \langle E_2, \sigma \rangle \Downarrow n_2}{\langle E_1 * E_2, \sigma \rangle \Downarrow n} \quad n = n_1 * n_2 \text{ (MULT)}$$

1.3 Semântica de Expressões Booleanas

$$\frac{}{\langle True, \sigma \rangle \Downarrow True} \text{ (TRUE)}$$

$$\frac{}{\langle False, \sigma \rangle \Downarrow False} \text{ (FALSE)}$$

$$\frac{\langle B, \sigma \rangle \Downarrow True}{\langle \neg B, \sigma \rangle \Downarrow False} \text{ (NOT1)}$$

$$\frac{\langle B, \sigma \rangle \Downarrow False}{\langle \neg B, \sigma \rangle \Downarrow True} \text{ (NOT2)}$$

$$\frac{\langle B_1, \sigma \rangle \Downarrow True}{\langle B_1 \vee B_2, \sigma \rangle \Downarrow True} \text{ (OR1)}$$

$$\frac{\langle B_1, \sigma \rangle \Downarrow False \quad \langle B_2, \sigma \rangle \Downarrow b}{\langle B_1 \vee B_2, \sigma \rangle \Downarrow b} \text{ (OR2)}$$

$$\frac{\langle B_1, \sigma \rangle \Downarrow True \quad \langle B_2, \sigma \rangle \Downarrow b}{\langle B_1 \wedge B_2, \sigma \rangle \Downarrow b} \text{ (AND1)}$$

$$\frac{B_1 \Downarrow False}{B_1 \wedge B_2 \Downarrow False} \text{ (AND2)}$$

$$\frac{\langle E_1, \sigma \rangle \Downarrow n_1 \quad \langle E_2, \sigma \rangle \Downarrow n_2}{\langle E_1 \leq E_2, \sigma \rangle \Downarrow b} \quad b = n_1 \leq n_2 \text{ (LEQ)}$$

1.4 Semântica de Comandos

$$\frac{}{\langle \text{Skip}, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma \rangle} \text{ (Skip)}$$

$$\frac{\langle E, \sigma \rangle \Downarrow n}{\langle x := E, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma[x \mapsto n] \rangle} \text{ (Atrib)}$$

$$\frac{\langle C_1, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma' \rangle \quad \langle C_2, \sigma' \rangle \Downarrow \langle \text{Skip}, \sigma'' \rangle}{\langle C_1 ; C_2, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma'' \rangle} \text{ (SEQ)}$$

$$\frac{\langle B, \sigma \rangle \Downarrow \text{True} \quad \langle C_1, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma' \rangle}{\langle \text{if } B \text{ then } C_1 \text{ else } C_2, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma' \rangle} \text{ (IF1)}$$

$$\frac{\langle B, \sigma \rangle \Downarrow \text{False} \quad \langle C_2, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma' \rangle}{\langle \text{if } B \text{ then } C_1 \text{ else } C_2, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma' \rangle} \text{ (IF2)}$$

$$\frac{\langle B, \sigma \rangle \Downarrow \text{True} \quad \langle C ; \text{While } B \text{ Do } C, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma' \rangle}{\langle \text{While } B \text{ do } C, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma' \rangle} \text{ (WHILE1)}$$

$$\frac{\langle B, \sigma \rangle \Downarrow \text{False}}{\langle \text{While } B \text{ do } C, \sigma \rangle \Downarrow \langle \text{Skip}, \sigma \rangle} \text{ (WHILE2)}$$